Simulink Design Verifier Report

lpa1_sldv

Anas

Simulink Design Verifier Report: lpa1_sldv

Anas

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Chapter 1. Summary

Analysis Information.

Model:	lpa1_sldv
Checksum:	
Mode:	Test generation
Test Generation Target:	Model
Status:	In progress
Elapsed Time:	2s

Objectives Status.

Number of Objectives:	**24**	
Objectives Satisfied:	22	(92%)
Objectives Undecided:	2	(8%)

Chapter 2. Analysis Information

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Model Information

File:	lpa1_sldv
Version:	1.4
Time Stamp:	Sat Jun 20 17:13:51 2026
Author:	Anas

Analysis Options

Mode:	TestGeneration
Rebuild Model Representation:	IfChangeIsDetected
Test Generation Target:	Model
Test Suite Optimization:	Auto
Maximum Testcase Steps:	10000time steps
Test Conditions:	UseLocalSettings
Test Objectives:	UseLocalSettings
Model Coverage Objectives:	ConditionDecision
Add tests for the missing coverage:	off
Include Relational Boundary Objectives:	off
Maximum Analysis Time:	300s
Block Replacement:	off
Parameters Analysis:	off
Include expected output values:	off
Randomize data that do not affect the outcome:	off
Additional analysis to reduce instances of rational approximation:	on
Save Harness:	off
Save Report:	off

User Artifacts

Coverage Data:	n/a
Test Data:	n/a

Chapter 3. Test Objectives Status

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Objectives Satisfied

Simulink Design Verifier generated test cases that exercise these test objectives.

#	Type	Model Item	Description	Analysis Time (sec)	Test Case
1	Condition	Relational Operator	RelationalOperator: input1 >= input2 **true**	-1	
2	Condition	Relational Operator	RelationalOperator: input1 >= input2 **false**	-1	
3	Condition	Relational Operator1	RelationalOperator: input1 <= input2 **true**	-1	
4	Condition	Relational Operator1	RelationalOperator: input1 <= input2 **false**	-1	
5	Condition	C1	Logic: input port 1 **true**	-1	
6	Condition	C1	Logic: input port 1 **false**	-1	
7	Condition	C1	Logic: input port 2 **true**	-1	
8	Condition	C1	Logic: input port 2 **false**	-1	
9	Decision	Abs	input < 0 **true**	-1	
10	Decision	Abs	input < 0 **false**	-1	
11	Condition	Relational Operator2	RelationalOperator: input1 <= input2 **true**	-1	
12	Condition	Relational Operator2	RelationalOperator: input1 <= input2 **false**	-1	
13	Condition	AND	Logic: input port 1 **true**	-1	
14	Condition	AND	Logic: input port 1 **false**	-1	

#	Type	Model Item	Description	Analysis Time (sec)	Test Case
15	Condition	AND	Logic: input port 2 **true**	-1	
16	Condition	AND	Logic: input port 2 **false**	-1	
17	Condition	AND	Logic: input port 3 **true**	-1	
18	Condition	AND	Logic: input port 3 **false**	-1	
19	Decision	Switch	logical trigger input **false (output is from 3rd input port)**	-1	
20	Decision	Switch	logical trigger input **true (output is from 1st input port)**	-1	
21	Decision	LPA controller/Saturation	input >= lower limit **true**	-1	
24	Decision	LPA controller/Saturation	input > upper limit **false**	-1	

Objectives Undecided

Simulink Design Verifier was not able to process these objectives with the current options.

#	Type	Model Item	Description	Analysis Time (sec)
22	Decision	LPA controller/Saturation	input >= lower limit **false**	-1
23	Decision	LPA controller/Saturation	input > upper limit **true**	-1

Chapter 4. Model Items

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This section presents, for each object in the model defining coverage objectives, the list of objectives and their individual status at the end of the analysis. It should match the coverage report obtained from running the generated test suite on the model, either from the harness model or by using the sldvruntest command.

Relational Operator

#:	Type	Description	Status	Test Case
1	Condition	RelationalOperator: input1 >= input2 true	Satisfied	n/a
2	Condition	RelationalOperator: input1 >= input2 false	Satisfied	n/a

Relational Operator1

#:	Type	Description	Status	Test Case
3	Condition	RelationalOperator: input1 <= input2 true	Satisfied	n/a
4	Condition	RelationalOperator: input1 <= input2 false	Satisfied	n/a

C1

#:	Type	Description	Status	Test Case
5	Condition	Logic: input port 1 true	Satisfied	n/a
6	Condition	Logic: input port 1 false	Satisfied	n/a

#:	Type	Description	Status	Test Case
7	Condition	Logic: input port 2 true	Satis-fied	n/a
8	Condition	Logic: input port 2 false	Satis-fied	n/a

Abs

#:	Type	Description	Status	Test Case
9	Decision	input < 0 true	Satis-fied	n/a
10	Decision	input < 0 false	Satis-fied	n/a

Relational Operator2

#:	Type	Description	Status	Test Case
11	Condition	RelationalOperator: in-put1 <= input2 true	Satis-fied	n/a
12	Condition	RelationalOperator: in-put1 <= input2 false	Satis-fied	n/a

AND

#:	Type	Description	Status	Test Case
13	Condition	Logic: input port 1 true	Satis-fied	n/a
14	Condition	Logic: input port 1 false	Satis-fied	n/a
15	Condition	Logic: input port 2 true	Satis-fied	n/a
16	Condition	Logic: input port 2 false	Satis-fied	n/a
17	Condition	Logic: input port 3 true	Satis-fied	n/a
18	Condition	Logic: input port 3 false	Satis-fied	n/a

Switch

#:	Type	Description	Status	Test Case
19	Decision	logical trigger input false (output is from 3rd input port)	Satisfied	n/a
20	Decision	logical trigger input true (output is from 1st input port)	Satisfied	n/a

LPA controller/Saturation

#:	Type	Description	Status	Test Case
21	Decision	input >= lower limit true	Satisfied	n/a
22	Decision	input >= lower limit false	Undecided	n/a
23	Decision	input > upper limit true	Undecided	n/a
24	Decision	input > upper limit false	Satisfied	n/a